

Session 4: Racialized Experiences in STEM

In this session, we will focus on understanding what being in STEM is like for Black, Indigenous, and other people of color (BIPOC). We will begin by reading an article that reframes the leaky pipeline as an unequal obstacle course, then listen to Dr. Ebony McGee discussing her personal experiences with racism in STEM. Our discussion will center on understanding racism and other forms of marginalization as systemic issues rather than interpersonal ones.

Things for facilitators to be prepared for:

Facilitators should be prepared for emotionally charged conversations about racism, privilege, and systemic inequities. It may be helpful to once again reference the community guidelines before having these conversations.

Participant Pre-Work:

- **Objective:** Introduce participants to the concept of racism in STEM and how those experiences take form.
 - **Activities:**
 - **Article:** [Nature Geoscience](#) - Scientists from Historically Excluded Groups Face a Hostile Obstacle Course
 - **Podcast:** [The Pitt Pulse Podcast Episode 1](#) - Introduction to Racism in STEM with Dr. Ebony McGee (31 min)
 - **Reflection 1:** Did the materials bring up any racialized experiences you have witnessed or lived?
 - If so, how did this experience feel?
 - **Reflection 2:** In the podcast episode, Dr. McGee states that mentoring programs “cannot dismantle 400 years of structural racism in society.” With that in mind, consider the following questions:
 - Is the work we are doing in this mentor training transformative?
 - If not, what purpose does it serve? What issues doesn't it solve?
 - What does being a culturally aware mentor do for mentees in your sphere of influence?
 - **Submission:** Be ready to share your questions and reflections during the synchronous session.
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Synchronous Session:

- **Materials needed:**
 - [Anonymous Writing Prompt slide](#)
 - [Session 4 Survey](#)
 - [Survey results from previous sessions](#)
 - Access to Zoom and familiarity with creating break-out rooms

- **Learning Objectives:**
 - **Objective 1:** Understand the experiences of marginalized groups in STEM and identify their obstacles to success.
 - **Objective 2:** Understand the difference between individual and systemic change.
- **Reviewing Survey Results:**
 - Before each session, all facilitators should plan to [review the survey results from the previous session](#). This is so they are able to incorporate any participant feedback into their session and make any adjustments.
- **Pework for Facilitator 2:**
 - Set all observers/facilitators as hosts (unless they want to be assigned to breakout rooms)
 - Set up screen with 5-minute Google timer, ready to share
 - **Facilitator 2** starts recording, **Facilitator 3** backs up at starting/stopping recording for breakouts
 - **Facilitator 3** starts recording.
- Note for all facilitators: make sure to rename yourself as “Facilitator” in the Zoom name.

Live Session Agenda:

- **Welcome and Warm-up Activity: 0:00 - 0:05 (5 min)**
 - **Activity Type:** Group Check-In
 - **Description:** **Facilitator 2** will paste into chat “As you enter, type in the chat. What do you think is one difference between systemic and interpersonal racism?”
- **Anonymous Writing Prompt 0:05 - 0:10 (5 min)**
 - **Description:** **Facilitator 1** explains the activity, and reads aloud the questions **Facilitator 2** puts in the chat. **Facilitator 2** shares a [Google Slide Link](#) in the chat where participants can anonymously share about racialized experiences in academia. As participants are writing, **Facilitator 2** selects 2-3 experiences for the group to discuss.
 - **Facilitator 2** shares instructions in the chat: “For the next few minutes, we will anonymously write our responses to the questions in one of the slides provided.
 - Did the pre-work bring up any racialized experiences you have witnessed or lived?
 - How did this experience feel?”
- **Group Discussion 0:10 - 0:25 (15 min)**
 - **Description:** **Facilitator 2** shares the slide that will be discussed. **Facilitator 1** reads one of the selected experiences out loud and guides participants through a discussion, also reading the guiding questions out loud as

they're posted in the chat. If time allows, the group can discuss a second and then third experience.

- **Facilitator 2** will post guiding questions to chat one at a time:
 - How would this experience have been different if the person involved had been white?
 - What were the effects of this experience on the person living it? Witnessing it?
 - How does this experience show elements of systemic racism?
- **BREAK: 00:25 - 00:30 (5 min)**
 - **Facilitator 2** shares their screen with the timer countdown and takes [attendance](#)
 - **Facilitator 3:** Stop recording during break, restarts recording after break
- **Group Discussion: 0:30-0:55 (25 min)**
 - **Description:** **Facilitator 1** will guide participants through an in-depth discussion about the utility of mentoring in the context of larger systemic issues in academia. **Facilitator 1** reads the guiding questions aloud as they're posted in the chat.
 - **Facilitator 2** posts these prompts to chat one at a time:
 - Is the work we are doing in this mentor training transformative?
 - If not, what purpose does it serve? What issues doesn't it solve?
 - What does being a culturally aware mentor do for mentees in your sphere of influence?
 - Why is it important to distinguish between individual and systemic change?
- **BREAK: 00:55 - 01:00 (5 min)**
 - **Facilitator 2** shares their screen with the timer countdown
 - **Facilitator 3:** Stop recording during break, start recording after break
- **Group Discussion: 01:00 - 01:25 (25 min)**
 - **Description:** **Facilitator 1** will guide participants through emotional processing and how to move forward. **Facilitator 1** reads the guiding questions aloud as they're posted in the chat.
 - **Facilitator 2** posts these questions to chat one at a time:
 - Individual efforts will never overturn systemic issues, and thousands of people have been working toward ending these systems for hundreds of years. When considering this, what emotions come up?
 - Is it possible to reframe this reality as hopeful? If so, how?
 - With this reality in mind, how do we move forward to address both systemic and individual issues in a mentorship role?
- **Wrap-Up: 01:25 - 01:30 (5 min)**
 - **Facilitator 1** summarizes key points from the session and opens the floor for final questions and comments.

- **Facilitator 2** shares the link to the [Session 4 Survey](#) in the chat and asks participants to complete it as soon as they can.
 - **Facilitator 1** ends the session by reminding participants of the pre-work for session 5, thanks them for joining, and ends the session.
 - **Facilitator 3** stops the recording.
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Facilitator Notes:

The Facilitators should spend time writing down everything they observed during the session (what worked, what didn't, and what improvements need to be made). They can also include advice for the next facilitator to watch out for. Facilitators may use the following guiding questions:

- How did the session go?
- Was there active participation in discussions?
- Do you feel like the learning objectives were met?
- Are there topics that need to be revisited in the next session?
- Were participants fully engaged when discussing their own mentoring experiences or goals?
- Were there times of vulnerability, disagreement, or hesitation? How were they managed?
- Did the conversation about group norms seem genuine and open? Were any important issues addressed?
- Are there any organizational improvements (tech, timing, breakout structure) to be addressed before the next session?